EfficientSAM: COCO Exploratory Data Analysis

Working draft | Updated 22 October 2026 | Completed analyses only

1. Dataset and scope

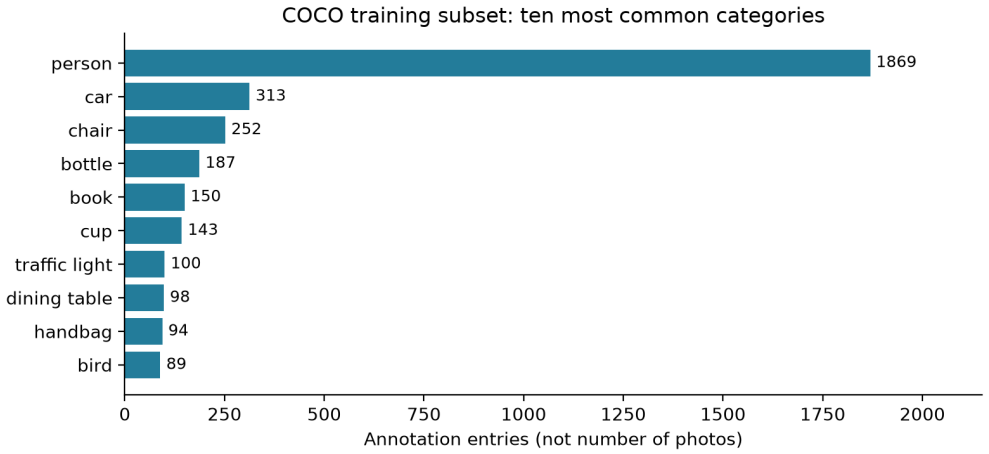
This course project uses official pretrained EfficientSAM checkpoints. The selected COCO 2017 pilot contains 800 images from the official training split and 200 from the official validation split, reserved for held-out evaluation. Sorted image metadata were sampled without replacement using Python random seeds 36219 and 36220 respectively. This random pilot is not guaranteed to represent every category or reproduce full COCO benchmark results.

The selected training images contain 5,916 annotation entries; the evaluation images contain 1,642. The user-run download checks reported that all 1,000 images decoded successfully and matched the dimensions in the metadata. Mask decoding and empty-mask checks are reported below; comprehensive annotation-geometry checks remain pending.

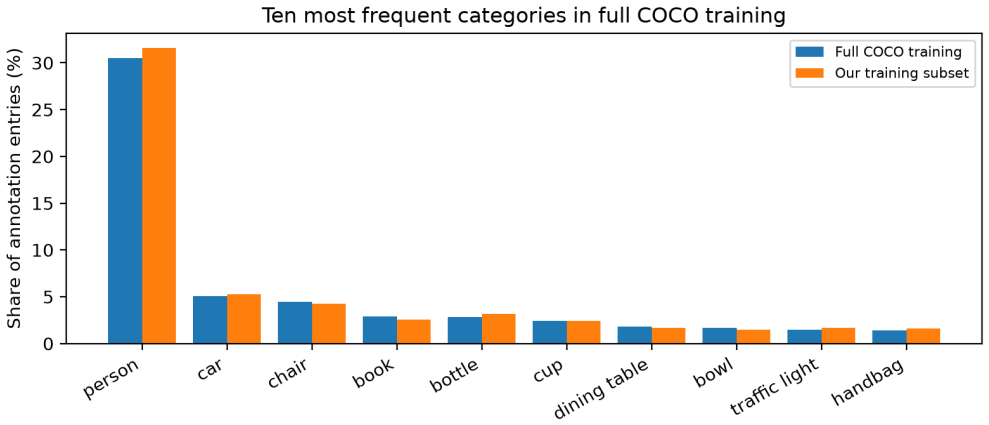
The training and evaluation subsets share 0 COCO image IDs. The two subsets are therefore disjoint by ID. Visual or near-duplicate similarity between different IDs has not been checked.

2. Training category distribution

People account for 1,869 of 5,916 training annotation entries (31.6%), compared with 313 car and 252 chair entries. The leading category therefore contributes substantially more annotations than the other categories shown. These are instance-annotation counts, not image counts; one image can contain several annotations of the same category.



The subset contains annotations for 79 categories. It includes 73 crowd annotation entries; these may describe groups rather than individual objects, so totals should not be interpreted as an exact count of individual physical objects.



Compared with 860,001 full training annotations, the subset broadly preserves the shares of these ten common categories. Person is 30.52% in full COCO versus 31.59% in the subset (+1.07 percentage points). This comparison does not establish rare-category coverage, representativeness of other visual attributes, or future model generalization.

3. Rare and absent training categories

The ten least frequent categories are toothbrush: 0; toaster: 1; microwave: 3; hair drier: 3; hot dog: 9; teddy bear: 9; stop sign: 14; bear: 14; broccoli: 14; snowboard: 15.

Toothbrush has no annotations in the training subset; toaster has only one. This describes annotation coverage, not proof that these objects are visually absent. Reliable category-specific conclusions cannot be drawn from such small counts. These coverage gaps must be considered when interpreting later evaluation results.

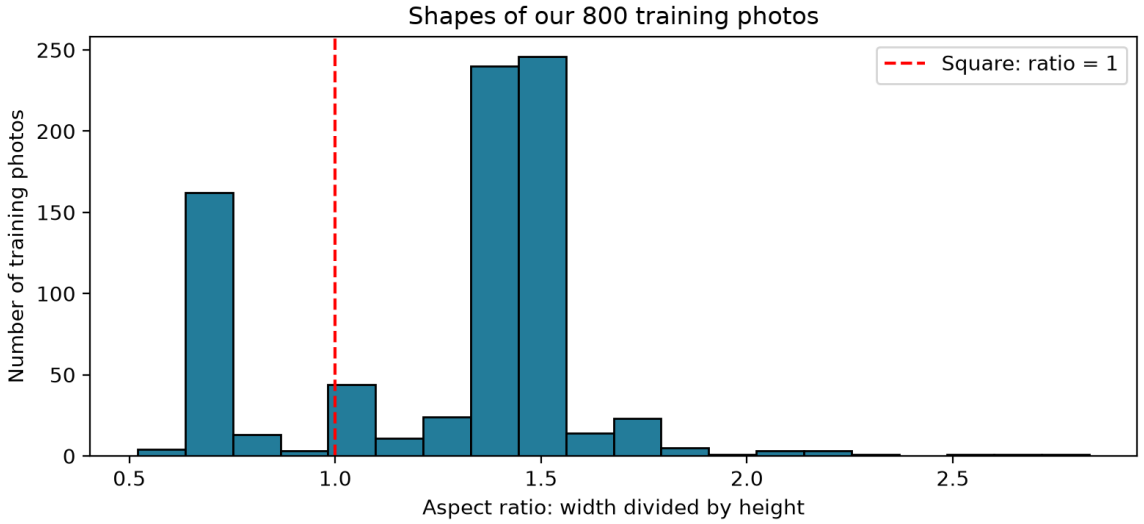
4. Evaluation category coverage

The 200 held-out evaluation images contain annotations for 76 of the 80 categories. Categories without annotations are fire hydrant, bear, snowboard, toaster. Per-category mask-overlap performance cannot be measured for these categories on this subset. Future results must report the evaluated category coverage and must not be described as full 80-category COCO benchmark results.

All 1,642 evaluation annotation masks were checked: 0 decoding failures and 0 empty masks. This confirms technical mask readability, not semantic correctness of every annotation.

5. Training image dimensions

Widths range from 300 to 640 pixels; heights range from 200 to 640 pixels. These are separate ranges and need not describe the same image. Width-to-height ratios range from 0.52 to 2.83, showing both portrait and landscape images. Direct square resizing can distort object shapes; later preprocessing must explicitly document its resizing policy and transform masks and prompt coordinates consistently.



The training subset contains 582 landscape images (ratio above 1), 183 portrait images (below 1), and 35 square images (equal to 1). Landscape images are most common. Histogram bars group nearby ratios; a bin crossing 1 should not be interpreted as an exact count of square images.

Planned evaluation metrics and their rationale

For a predicted foreground mask P and ground-truth mask G, IoU = |P intersection G| / |P union G|, and Dice = 2|P intersection G| / (|P| + |G|). Both range from 0 (no overlap) to 1 (perfect overlap). Mean foreground IoU over eligible non-crowd object instances is the proposed primary score, with Dice as a companion. These related overlap scores do not constitute independent evidence. Pixel accuracy is unsuitable as the primary score because a mostly-background prediction can score highly while missing small objects.

Report point-prompt and box-prompt results separately under fixed prompt-generation rules. Include category and object-size breakdowns, their sample counts, and macro category averages over categories with eligible examples. Exclude crowd regions from the primary single-instance protocol and report their count. Subset overlap scores are not full COCO mask AP. These are proposed measurements, not model results; model evaluation has not begun.

6. Raw visual example and object coverage

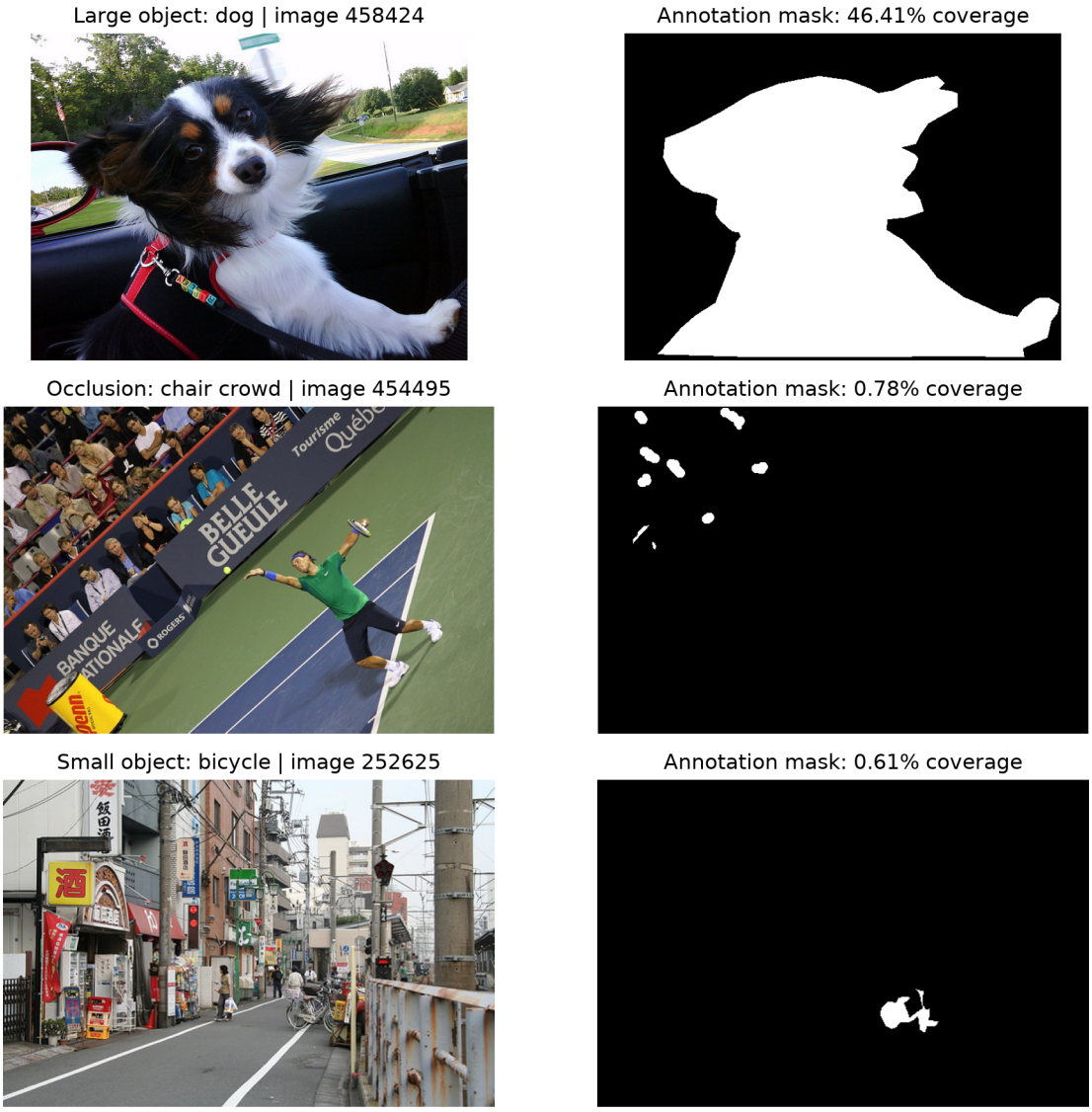


Figure: Three deliberately selected training examples, with original photos on the left and decoded annotation masks on the right. White indicates foreground. These illustrate different challenges and are not a random sample of the dataset.

The first training annotation is a dog in a 640 x 480 image. Its stored segmentation area is 142,495.78 square pixels, covering 46.39% of the image. Coverage is annotation area divided by image width times height, multiplied by 100. This is one illustrative object, not a dataset-wide summary. Stored polygon-based areas can be fractional and may differ slightly from decoded binary-mask pixel counts.

The chair crowd mask contains scattered visible regions in a scene with occlusion. The selected bicycle covers only 0.61% of its street photo, illustrating small-scale detail amid background clutter. The food-truck anomaly is shown on the next page.

Image 458424 source: http://farm3.staticflickr.com/2784/4123723564_47ce7cb930_z.jpg. License: http://creativecommons.org/licenses/by-nc-nd/2.0/

Image 454495 source: http://farm7.staticflickr.com/6184/6082882896_e01b39e1e9_z.jpg. License: http://creativecommons.org/licenses/by-nc-sa/2.0/

Image 252625 source: http://farm1.staticflickr.com/46/188068269_52967f417f_z.jpg. License: http://creativecommons.org/licenses/by-nc-nd/2.0/

7. Area anomaly and mask verification

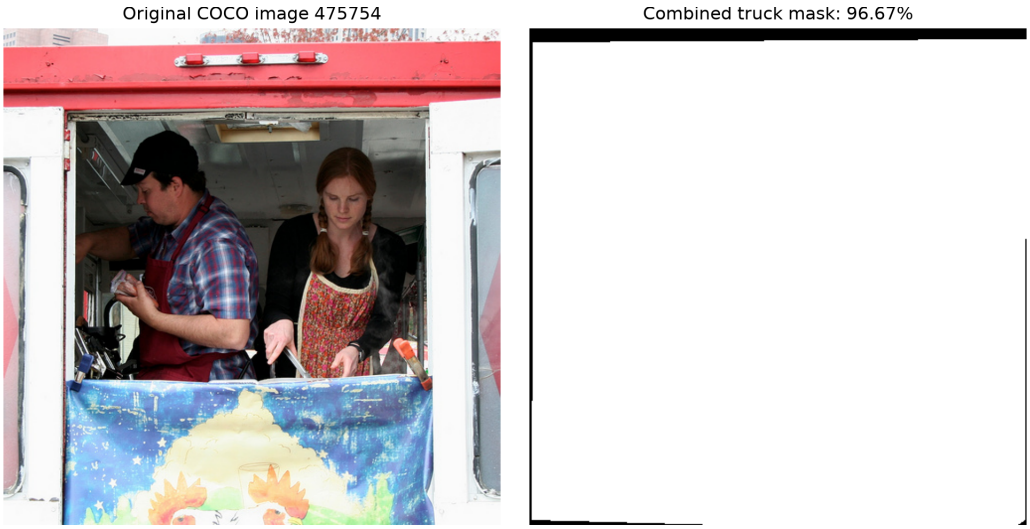
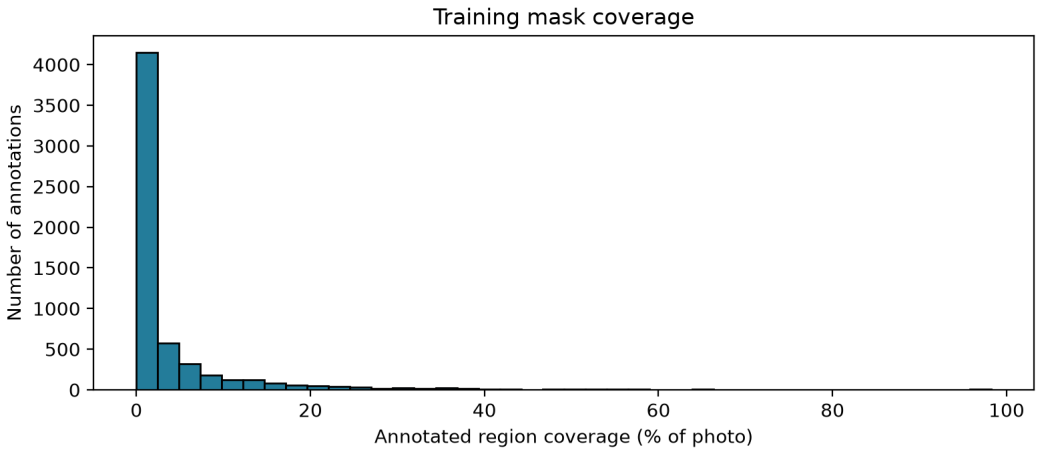


Figure: Close-up of a food-truck service window (left) and the combined mask for its truck annotation (right). White pixels belong to the annotated region. The original photo is displayed without an overlay; the mask is a separate panel.

Truck annotation 397929 in image 475754 has stored area 336,338.72 in a 500 x 500 image, implying 134.54% coverage. Its two polygons lie within image bounds but overlap. The official COCO mask toolkit gives individual rasterized areas of 241,673 and 94,666 pixels; their union is 241,673 pixels (96.67% coverage). The stored area closely matches the sum of the polygon areas rather than their union. For this example, overlapping area explains the inflated value. Subsequent coverage analysis uses decoded union masks; the training-wide check follows. Source annotations are preserved unchanged.

Training-wide mask check

All 5,916 training annotation entries were decoded with COCO annToMask; 0 failed, and 0 decoded masks are empty. Each training annotation therefore yields at least one foreground pixel. Coverage measured from binary foreground pixels ranges from 0.0004% to 98.24%. Median coverage is 0.64%, placing the middle annotation below 1% of image area and demonstrating that small relative regions are common. Thus the subset includes very small regions and regions occupying almost an entire image. Small regions may lose detail during downsampling. Successful decoding confirms that masks can be constructed; it does not establish that every annotation is semantically correct. These results cover training annotations only, including crowd entries.



The histogram is right-skewed: most annotation regions occupy a small fraction of image area, with a long tail of larger regions. A bin near zero spans a range of small coverages; its height does not mean those masks have zero area.

Image source recorded in COCO: http://farm3.staticflickr.com/2680/4465762706_ee94b80f00_z.jpg. License: Attribution-NoDerivs 2.0 (https://creativecommons.org/licenses/by-nd/2.0/).

Source: COCO 2017 instance annotations, https://cocodataset.org/#download. Counts computed from saved subsets; image-check results supplied by the user.